Problem 1 :

Partie a) Montrer que le champ de vecteurs : $\vec{F} = \langle e^x yz, e^x z + 2yz, e^x y + y^2 + 1\rangle$ est conservatif

$\vec{F}$ est conservatif si $\text{curl}\,\vec{F} = 0$ (dans un domaine simplement connexe)

$\text{curl}\,\vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix} \Leftrightarrow (R_y - Q_z)\hat{\imath} + (P_z - R_x)\hat{\jmath} + (Q_x - P_y)\hat{k}$

Ici : $P = e^x yz$ $\quad R_y - Q_z = (e^x + 2y) - e^x - 2y = 0$

$Q = e^x z + 2yz$ $\quad P_z - R_x = e^x y - e^x y = 0$

$R = e^x y + y^2 + 1$ $\quad Q_x - P_y = e^x z - e^x z = 0$

$\text{curl}\,\vec{F} = 0\hat{\imath} + 0\hat{\jmath} + 0\hat{k} = \vec{0}$ partout, $\vec{F}$ est bien conservatif

Partie b) Trouver un potentiel tel que $\nabla f = \vec{F}$

$f_x = P = e^x yz \Leftrightarrow e^x yz + \overset{\partial g/\partial y}{g(y,z)} = f$

$f_y = Q = e^x z + 2yz \Leftrightarrow e^x z + g'(y,z)$

$e^x z + 2yz = e^x z + g'(y,z) \Leftrightarrow g'(y,z) = 2yz \Leftrightarrow g = y^2 z + h(z) \quad (\Rightarrow f = e^x yz + y^2 z + h(z)$

$f_z = R = e^x y + y^2 + 1 \Leftrightarrow \frac{\partial f}{\partial z} = e^x y + y^2 + h'(z) \Leftrightarrow h'(z) = 1 \Leftrightarrow h(z) = z + C$

$f(x,y,z) = e^x yz + y^2 z + z + C$

Partie c) Montrer que $\vec{G} = \langle y, x, y\rangle$ n'est pas conservatif

$P = y, Q = x, R = y$

$R_y - Q_z = 1 - 0 \quad \text{curl}\,\vec{F} = 1\hat{\imath} + 0\hat{\jmath} + 0\hat{k}$

$P_z - R_x = 0 - 0 \quad \text{curl}\,\vec{F} \neq 0$

$Q_x - P_y = 1 - 1$

Problem 2 :

Partie a) Calculer le flux sortant à travers S du champ $\vec{F} = y\hat{\imath} + x\hat{\jmath} + z\hat{k}$

Partie "horizontale" du flux $= \langle y, -x, 0\rangle \cdot \langle x, y, z\rangle = 0$

La partie "rotationnelle" $\langle y, -x, 0\rangle$ de $\vec{F}$ est tangente à la sphère

Flux: $\vec{F} \cdot \hat{n} = \langle y, -x, z\rangle \cdot \frac{\langle x, y, z\rangle}{2} = \frac{yx - xy + z^2}{2} = \frac{z^2}{2}$

$\iint_S \frac{z^2}{2}\, dS$

Passage en coordonnées sphériques :

$a = 2 \quad dS = a^2 \sin\phi\, d\phi\, d\theta = 4\sin\phi\, d\phi\, d\theta$ et $z = a\cos\phi = 2\cos\phi$

$x^2 + y^2 = a^2 \sin^2\phi = 4\sin^2\phi$ $\quad$ On veut $4\sin^2\phi = 1 \Leftrightarrow 2\sin\phi = 1 \Leftrightarrow \sin\phi = \frac{1}{2} \Leftrightarrow \phi = \begin{cases} \frac{\pi}{6} \\ \frac{5\pi}{6} \end{cases}$

$\text{Flux} = \iint_S \frac{z}{2}\, dS = \int_0^{2\pi} \int_{\pi/6}^{5\pi/6} \frac{(2\cos\phi)^2}{2}\, 4\sin\phi\, d\phi\, d\theta$

$= \int_0^{2\pi} \int_{\pi/6}^{5\pi/6} 8\cos^2\phi\, \sin\phi\, d\phi\, d\theta$

$\cos^2\phi \sin\phi$ on pose $u = \cos\phi \Leftrightarrow u^2 du = \cos^2\phi \sin\phi$

$\frac{du}{d\phi} = -\sin\phi \Leftrightarrow du = -\sin\phi\, d\phi \Leftrightarrow \sin\phi\, d\phi = (-du)$

Avec $u = \cos\phi$: $\int \cos^2\phi \sin\phi\, d\phi = \int u^2(-du) = -\int u^2 du \Leftrightarrow -\int \frac{u^3}{3} \Leftrightarrow \int \frac{-\cos^3\phi}{3}$

$\int_{\pi/6}^{5\pi/6} 8\cos^2\phi \sin\phi\, d\phi = 8\left[\frac{-\cos^3\phi}{3}\right]_{\pi/6}^{5\pi/6} = \frac{-8}{3}\left[\cos^3\phi\right]_{\pi/6}^{5\pi/6} = -\frac{8}{3}\left[\frac{-3\sqrt{3}}{8} - \frac{3\sqrt{3}}{8}\right] \Leftrightarrow \frac{-8}{3} \times \frac{-3\sqrt{3}}{4} = 2\sqrt{3}$

$\cos(5\pi/6) = -\frac{\sqrt{3}}{2} \quad \left(-\frac{\sqrt{3}}{2}\right)^3 = -\frac{3\sqrt{3}}{8}$

$\cos(\pi/6) = \frac{\sqrt{3}}{2} \quad \left(\frac{\sqrt{3}}{2}\right)^3 = \frac{3\sqrt{3}}{8}$

$\text{Flux} = \int_0^{2\pi} 2\sqrt{3}\, d\theta \Leftrightarrow 2\sqrt{3}\int_0^{2\pi} \theta \Leftrightarrow 2\sqrt{3} \times 2\pi = 4\sqrt{3}\pi$

Partie b : Montrer que le flux de $\vec{F} = \langle y, x, z\rangle$ à travers n'importe quelle portion de la surface cylindrique est nul.

Sur la surface du cylindre $x^2 + y^2 = 1$, le vecteur normal sortant est horizontal $\hat{n} = \langle x, y, 0\rangle$, rayon $= 1 \Leftrightarrow \langle x, y, 0\rangle = 1$

$\vec{F} \cdot \hat{n} = \langle y, -x, z\rangle \cdot \langle x, y, 0\rangle$

$\vec{F} \cdot \hat{n} = yx - xy + z \times 0 = 0$

Partie c) En utilisant le théorème de la divergence appliqué à $\vec{F}$, calculer le volume de la région entre S et le cylindre.

$\text{div}\,\vec{F}$: $P_x = 0, Q_y = 0, R_z = 1 \quad \text{div}\,\vec{F} = 0 + 0 + 1 = 1$

Donc : $\iint_S \vec{F} \cdot \hat{n}\, dS = \iiint_D \text{div}\,\vec{F}\, dV = \iiint_D 1\, dV = \text{Volume}(D)$

Deux parties : $\underbrace{S}_{\text{flux} = 4\sqrt{3}\pi}$ et la portion de cylindre (flux $= 0$), Flux total $= 4\sqrt{3}\pi + 0 = 4\sqrt{3}\pi$

Problem 3 :

Partie a) $z = 1 \Leftrightarrow x^2 + y^2 + z^2 = 2 \Leftrightarrow x^2 + y^2 + 1^2 = 2 \Leftrightarrow x^2 + y^2 = 2 - 1 \Leftrightarrow x^2 + y^2 = 1$

C est un cercle de rayon 1

On paramétrise : $x = \cos t$; $y = \sin t$; $z = 1$ avec $t \in [0; 2\pi]$

$dx = -\sin t\, dt$ $\quad \vec{F} \cdot d\vec{r} = xz\,dx + y\,dy + y\,dz \Leftrightarrow \vec{F} \cdot d\vec{r} = \cos t \times (-\sin t\, dt) + \sin t \times (\cos t\, dt) + \sin t \times 0$

$dy = \cos t\, dt$ $\quad \Leftrightarrow \vec{F} \cdot d\vec{r} = 0\, dt$

$dz = 0\, dt$

$P = xz = \cos t$

$Q = y = \sin t$ $\quad$ Donc : $\int_0^{2\pi} 0\, dt = 0$

$R = y = \sin t$

Partie b) Calculer curl $\vec{F}$ pour $\vec{F} = xz\hat{\imath} + y\hat{\jmath} + y\hat{k}$

$P = xz$; $Q = y$; $R = y$ $\quad \text{curl}\,\vec{F} = (R_y - Q_z)\hat{\imath} + (P_z - R_x)\hat{\jmath} + (Q_x - P_y)\hat{k}$

$\text{curl}\,\vec{F} = (1 - 0)\hat{\imath} + (x - 0)\hat{\jmath} + (0 - 0)\hat{k}$

$\text{curl}\,\vec{F} = \hat{\imath} + x\hat{\jmath}$

Partie c) $\underbrace{\oint \vec{F} \cdot d\vec{r}}_{} = \iint_S \text{curl}\,\vec{F} \cdot \hat{n}\, dS$

$I = \iint_S \text{curl}\,\vec{F} \cdot \hat{n}\, dS$

$0 = \iint_S \langle 1, x, 0\rangle \cdot \hat{n}\, dS$

Problem 4 : $\vec{F} = \hat{\imath} + \hat{\jmath} + \hat{k} \qquad x^4 + y^4 + z^4 = 1$

$\text{div}\,\vec{F} = 0 \Leftrightarrow \iint_S \vec{F} \cdot \hat{n}\, dS = \iiint_D \text{div}\,\vec{F}\, dV = \iiint_D 0\, dV = 0$

Le flux sortant est nul.

Problem 5 : surface $S = z = (x^2 + y^2 + z^2)^2$

a) Somme de carré élevée au carré est forcément positive ou nulle, donc $z \geq 0$ pour tout point de S

b) $z = \rho\cos\phi \quad x^2 + y^2 + z^2 = \rho^2$

$\Leftrightarrow z = (x^2 + y^2 + z^2)^2$

$\Leftrightarrow \rho\cos\phi = (\rho^2)^2 \Leftrightarrow \rho\cos\phi = \rho^4 \Leftrightarrow \cos\phi = \rho^3$

$\Leftrightarrow \rho = \sqrt[3]{\cos\phi}$

c) $V = \int_0^{2\pi} \int_0^{\pi/2} \int_0^{(\cos\phi)^{1/3}} \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$

18.02 – Practice Exam 4A

Problem 1. (15)

a) Show that the vector field

$$\mathbf{F} = \langle e^x yz, e^x z + 2yz, e^x y + y^2 + 1\rangle$$

is conservative.

b) By a systematic method, find a potential for **F**.

c) Show that the vector field $\mathbf{G} = \langle y, x, y\rangle$ is not conservative.

Problem 2. (20) Let S be the part of the spherical surface $x^2 + y^2 + z^2 = 4$, lying in $x^2 + y^2 > 1$, which is to say outside the cylinder of radius one with axis the z-axis.

a) Compute the flux outward through S of the vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + z\mathbf{k}$.

b) Show that the flux of this vector field through any part of the cylindrical surface is zero.

c) Using the divergence theorem applied to **F**, compute the volume of the region between S and the cylinder.

Problem 3. (20) Let S be the part of the spherical surface $x^2 + y^2 + z^2 = 2$ lying in $z > 1$. Orient S upwards and give its bounding circle, C, lying in $z = 1$ the compatible orientation.

a) Parametrize C and use the parametrization to evaluate the line integral

$$I = \oint_C xz\,dx + y\,dy + y\,dz.$$

b) Compute the curl of the vector field $\mathbf{F} = xz\mathbf{i} + y\mathbf{j} + y\mathbf{k}$.

c) Write down a flux integral through S which can be computed using the value of I.

Problem 4. (15) Use the divergence theorem to compute the flux of $\mathbf{F} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ outwards across the closed surface $x^4 + y^4 + z^4 = 1$.

Problem 5. (15) Consider the surface S given by the equation

$$z = (x^2 + y^2 + z^2)^2.$$

a) Show that S lies in the upper half space ($z \geq 0$).

b) Write out the equation for the surface in spherical polar coordinates.

c) Using the equation obtained in part b), give an iterated integral, with explicit integrand and limits of integration, which gives the volume of the region inside this surface. Do not evaluate the integral.

Problem 6. (15) Let S be the part of the surface $z = xy$ where $x^2 + y^2 < 1$. Compute the flux of $\mathbf{F} = y\mathbf{i} + x\mathbf{j} + z\mathbf{k}$ upward across S by reducing the surface integral to a double integral over the disk $x^2 + y^2 < 1$.

Problem 6 : $z = xy \qquad x^2 + y^2 < 1 \qquad \vec{F} = y\hat{\imath} + x\hat{\jmath} + z\hat{k}$

$z = f(x,y) : \hat{n}\, dS = \langle -f_x, -f_y, 1\rangle\, dx\,dy$

Ici : $\hat{n}\, dS = \langle -y, -x, 1\rangle\, dx\,dy$

$\vec{F} \cdot \hat{n}\, dS = \langle y, x, xy\rangle \cdot \langle -y, -x, 1\rangle\, dx\,dy$

$\Leftrightarrow -y^2 - x^2 + xy\, dx\,dy$

$\text{flux} = \iint_{x^2 + y^2 < 1} (-x^2 - y^2 + xy)\, dx\,dy$

Coordonnées polaires : $x = r\cos\theta, y = r\sin\theta, dx\,dy = r\,dr\,d\theta$

$xy = r\cos\theta\, r\sin\theta \Leftrightarrow r^2\cos\theta\sin\theta$

$x^2 + y^2 = r^2$

$\text{flux} = \int_0^{2\pi} \int_0^1 (-r^2 + r^2\cos\theta\sin\theta)\, r\,dr\,d\theta$

La partie $\int_0^{2\pi} \cos\theta\sin\theta\, d\theta = 0$ donc inutile

$\text{flux} = \int_0^{2\pi} \int_0^1 (-r^2) \cdot r\,dr\,d\theta = \int_0^{2\pi} \int_0^1 -r^3\, dr\,d\theta$

$\Leftrightarrow \int_0^{2\pi} \left[-\frac{r^4}{4}\right]_0^1 d\theta \Leftrightarrow \int_0^{2\pi} -\frac{1}{4}\, d\theta \Leftrightarrow \int_0^{2\pi} -\frac{\theta}{4} \Leftrightarrow \frac{-2\pi}{4} = -\frac{\pi}{2}$